

FinWise Login System – Setup & Working Documentation

Overview

This document explains how to set up and run the FinWise Login and Authentication Module, which includes:

- User login and signup using MongoDB
- Password hashing with bcrypt
- Google Sign-In with Firebase
- Forgot Password via OTP verification
- Frontend: React.js
- Backend: Flask (Python)
- Database: MongoDB

System Requirements

Component	Version / Details
Python	3.13.x
Node.js	18+
MongoDB	v8.0+
Firebase Console	Access required for Google login
OS	Windows 10 / 11

MongoDB Setup

Database Info:

Your MongoDB is already set up with:

- Database Name: th3ee
- Collection Name: users

Example document:

```
{  
  "_id": "ObjectId('...')",  
  "name": "Test User",  
  "phone": "+911234567890",  
  "password": "Encrypted bcrypt hash",  
  "cart": []  
}
```

Connection Settings:

```
client = MongoClient("mongodb://localhost:27017/")  
db = client["th3ee"]  
users = db["users"]
```

Starting MongoDB

Step 1: Run as Administrator

```
net start MongoDB
```

Step 2: Verify

```
mongod --version
```

```
mongo
```

```
use th3ee
```

```
show collections
```

Flask Backend Setup

1. Create Virtual Environment:

```
python -m venv venv
```

```
venv\Scripts\activate
```

2. Install Dependencies:

`pip install flask flask-cors pymongo bcrypt firebase-admin`

3. Run Flask App:

`python app.py`

Runs on `http://127.0.0.1:5001`

Google Login Setup

1. Configure Firebase:

- Go to `console.firebase.google.com`
- Create project FinWise
- Enable Google Sign-in
- Add Web App and get Firebase config

2. Download Service Account:

- Go to Project Settings → Service Accounts
- Generate new private key
- Save as `serviceAccountKey.json` in backend folder

3. Backend verifies Google login token through `firebase_admin.verify_id_token(id_token)`

Already done above thing!! Here is the link of firebase console:

<https://console.firebase.google.com/u/0/project/finwise-cfa37/settings/general/web:MmE3MGlyMzktOThiMi00ZmRkLWFhMmMtNjY2YTA1ODE0ZDYx>

Forgot Password (OTP Reset)

Endpoints:

`/send-otp` - Generates and sends OTP

/verify-otp - Validates OTP

/forgot-password - Updates password securely

In dev mode, OTP printed in console:

[DEV OTP] phone=+911234567890 otp=123456

Frontend Setup (React)

1. Install Dependencies:

npm install

2. Start Frontend:

npm start

Runs on http://localhost:3000

Connection Summary

Component	Port	Description
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Flask Backend	5001	Handles authentication APIs
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React Frontend	3000	UI for user login/signup
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MongoDB	27017	Database for users
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Testing Steps

Signup → Creates user in MongoDB

Login → Verifies hashed password

Continue with Google → Creates user via Firebase token

Forgot Password → Sends OTP and updates password

Backend Routes Summary

Route	Method	Description
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/signup	POST	Create new user
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/login	POST	Login with credentials
/google-login	POST	Google Sign-In
/send-otp	POST	Send OTP for password reset
/verify-otp	POST	Verify entered OTP
/forgot-password	POST	Update password

Result

The FinWise login module supports:

- Secure local authentication
- Google Sign-In integration
- OTP-based password reset
- Real-time MongoDB connectivity